Transient Structure as a Sparse-Control Signal: Matched-Budget Graph Selection Before Continual Learning

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Abstract

Transformer inference repeatedly constructs relational statistics that disappear after the request. Before consolidating them into persistent state, we ask whether compact graph features select sparse memory better than score-only rules at matched materialization budgets. A controlled weak-bridge suite establishes mechanism sufficiency: at four chunks, bridge preservation raises quality from 0.297 to 0.627, while a validation-fitted structural selector reaches a generator-specific ceiling. We then close the candidate-level natural-data gap with frozen Qwen3-0.6B on identity-disjoint HotpotQA and QASPER examples. Twelve 32-token candidates are evaluated at 64, 128, and 192 native K/V tokens; every condition performs actual causal K/V prefill and answer-likelihood scoring. Bridge preservation has positive mean answer-logprob deltas in all six dataset/budget cells, from +0.011 to +1.086, but every paired 95% bootstrap interval includes zero. The validation-fitted combined selector is mostly negative, and structural candidate-removal models retain negative held-out R^2. Natural bridge structure is therefore suggestive but not reproducible at this diagnostic scale. The gate for online consolidation remains closed.

1 Question and Evidence Gate

Let $C = \{c_1, \ldots, c_n\}$ be candidate chunks with base discovery scores s_i. The question is whether transient structure supplies incremental information about which candidates should be materialized while model weights remain frozen. This is not the claim that training ignores attention: gradients already pass through attention. The narrower hypothesis is that recurrent sample-level structure may provide a useful sparse-control statistic that is normally discarded after inference.

The experiment separates observation, intervention, and learning. Observation constructs compact candidate features. Intervention changes only frozen candidate selection. Learning of prototypes, adapters, or model weights is prohibited in this paper. Online consolidation may begin only after either (i) structural features predict held-out causal candidate utility beyond score and surface controls, or (ii) a structural selector reproducibly shifts the quality–materialization Pareto frontier on natural tasks.

2 Relation to Papers 0–0.6

Paper 0 defines Neural Structural Consolidation and its stop/go criteria. Paper 0.5 supplies recurrent n-gram controls, exact trace locations, and intervention discipline. Paper 0.6 shows weak hierarchy geometry but no semantic attention-motif specificity. Paper 1 therefore asks whether transient graph statistics have *operational* value before treating them as a substrate for continual learning. Geometry, attention weight, and probe performance alone are not accepted as mechanistic evidence.

3 Frozen Sparse-Control Formalization

Materializing $M \subseteq C$ incurs

$$B(M) = \sum_{c_i \in M} |KV(c_i)|.$$

Every primary comparison receives the same candidates and exact chunk budget. For candidate i, the compact feature vector is

$$x_i = [s_i, L_i, S_i, E_i, P_i, A_i, C_i, G_i],$$

where L and S are lexical and semantic discovery controls, E is entropy contribution, P is cross-layer/head persistence, A is rank agreement, C is within-community centrality, and G is cross-community bridge mass. No token-level N^2 attention matrix is retained in traces.

For a diagnostic set with known evidence, removal utility is

$$U_i = Q(C) - Q(C \setminus \{c_i\}).$$

We compare ridge prediction from (s, L, S) against (s, L, S, E, P, A, C, G) on held-out examples. Selector S5 is stricter: it may combine only s with (E, P, A, C, G), so it cannot silently reweight lexical or semantic discovery channels.

4 Implemented Conditions

ID	Condition
B0	Base score with exact top-k.
B1	Budget-tuned top-k evaluated over the same full budget sweep.
B2	Fixed power sharpening $p_i' \propto p_i^{\gamma}$.
B3	Entropy-adaptive global sharpening.
S1	Base score plus cross-layer/head persistence.
S2	Base score plus rank agreement.
S3	Base score plus within-community centrality.
S4	Explicit reservation for high-bridge candidates.
S5	Validation-fitted base-plus-structure linear selector.
O1	Oracle evidence ordering, diagnostic upper bound only.

Table 1: Implemented frozen selectors. B0–B3 and S1–S5 obey exact candidate and materialization budgets.

A global monotone transform cannot change exact top-k membership. Consequently B2 and B3 must reproduce B0 under a fixed cardinality budget; this is an algebraic control rather than a failed implementation. Sharpening can matter only with thresholds, variable budgets, sampling, or altered model-native attention, none of which is conflated with this selector test.

5 Controlled Weak-Bridge Suite

Each example contains 24 candidates partitioned into three communities, with two- or three-hop evidence paths, matched 32-token chunks, high/low lexical-overlap regimes, three bridge strengths, six synthetic layers, and four synthetic heads. Evidence candidates receive noisy semantic support, but the indispensable cross-community bridge is down-weighted in the base score and a lexical distractor is strengthened. Validation and test examples have disjoint deterministic identities. Five seeds each contribute 80 validation and 120 test examples, for 1,000 total and 600 held-out scientific units.

Quality combines evidence recall and complete-path recovery:

$$Q(M) = \tfrac{1}{2}\frac{|M \cap E|}{|E|} + \tfrac{1}{2}\mathbf{1}[E \subseteq M].$$

Budgets are $k \in \{2, 3, 4, 6, 8\}$ chunks. Reported Qwen-equivalent K/V bytes use eight K/V heads, head width 128, two tensors, and two bytes per element, or 4096 bytes per materialized token. These are accounting-equivalent bytes, not a measured native materialization run.

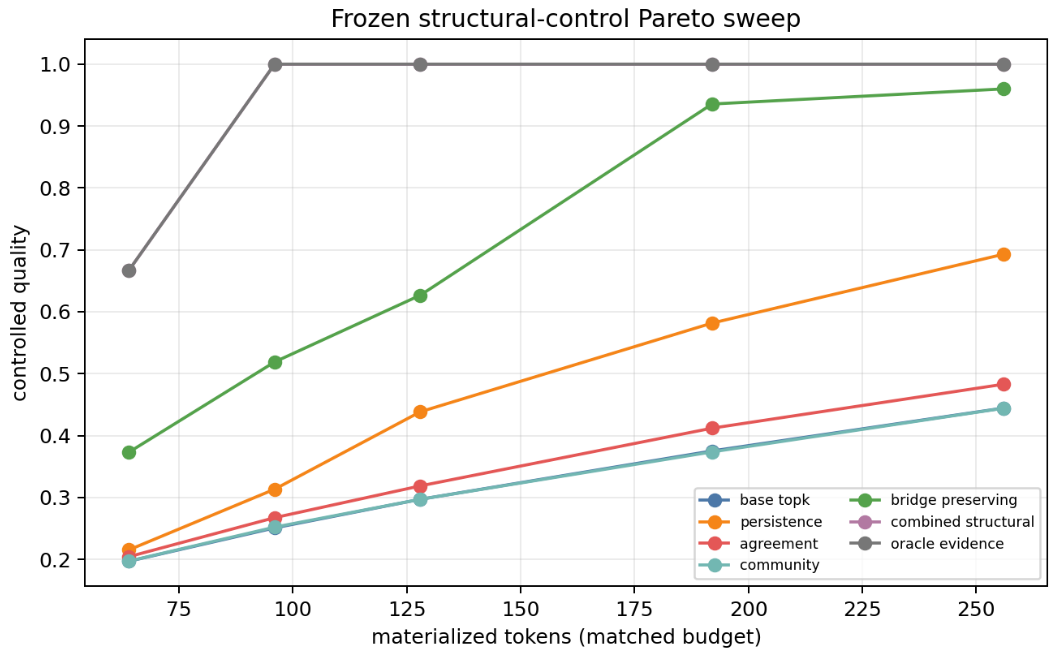

Figure 1: Controlled quality versus exactly matched materialized tokens. The oracle is diagnostic. Structural gains occur because the generator contains a known cross-community bridge mechanism.

6 Frozen-Qwen Residualized Audit

The available PRA artifact retains layerwise graph summaries, but intentionally omits its 7.8 GB candidate tensor cache. We therefore perform an observational audit rather than reconstructing a candidate-selection result. The source contains 672 rows from 84 identity-disjoint 2WikiMultiHopQA and MuSiQue examples, each measured at layers $0, 4, 8, 12, 16, 20, 24, 27$ of frozen Qwen3-0.6B revision `c1899de...`. Validation identities fit the models; test identities evaluate them.

The target is the fraction of layers with complete native graph recovery, aggregated to one row per example before evaluation. Surface controls are candidate/parent count, annotated hops, graph density, layer, dataset, and graph type. Added structural predictors are duplicate-neighbor rate, reciprocal-edge rate, effective branching factor, unique neighbors, shortcut rate, giant-component fraction, and mean out-degree. This analysis is predictive and identity-disjoint, but not causal: it has neither candidate removal outcomes nor task-generation quality.

7 Natural Candidate-Level Intervention

We regenerate candidate features directly from frozen Qwen3-0.6B revision `c1899de...`. The diagnostic slice contains eight validation and eight test identities from each of HotpotQA and QASPER. Hotpot identities are examples; QASPER identities are papers, with at most one question per paper, so no paper crosses the split. Each example supplies twelve fixed 32-token candidates. Five depth samples and four channel groups produce compact query–candidate scores and candidate similarities; traces retain derived persistence, agreement, communities, and bridge mass, never full attention matrices.

Selection uses budgets of 64, 128, and 192 tokens. Selected tokens are passed through the model's native causal prefill with `use_cache`; K/V bytes are counted from the resulting tensors. These budgets materialize 7.34, 14.68, and 22.02 MB respectively. The primary quality endpoint is mean target-answer token log probability under the frozen model. Evidence recall and complete evidence recovery are diagnostics. Candidate-removal utility is computed on one predeclared validation and test example per dataset because each candidate requires another native forward pass.

8 Results

Condition	Quality	Evidence recall	Path complete	Bridge recall
base topk	0.297	0.526	0.068	0.102
power sharpen	0.297	0.526	0.068	0.102
persistence	0.438	0.657	0.220	0.262
agreement	0.319	0.551	0.087	0.122
community	0.297	0.526	0.068	0.102
bridge preserving	0.627	0.774	0.480	0.785
combined structural	1.000	1.000	1.000	1.000
oracle evidence	1.000	1.000	1.000	1.000

Table 2: Controlled test performance at four 32-token chunks. The oracle is diagnostic only. All non-oracle conditions receive identical candidates and budgets.

At four chunks, explicit bridge preservation increases quality by 0.330 and complete-path rate by 0.412 over B0. Persistence also helps, but agreement is small and community centrality alone is identical to B0 at this operating point. S5 reaches the oracle ceiling from three chunks upward and also matches its quality at two chunks. This ceiling is a generator diagnostic: validation and test share the same structural mechanism, so it is not evidence of natural-task generalization.

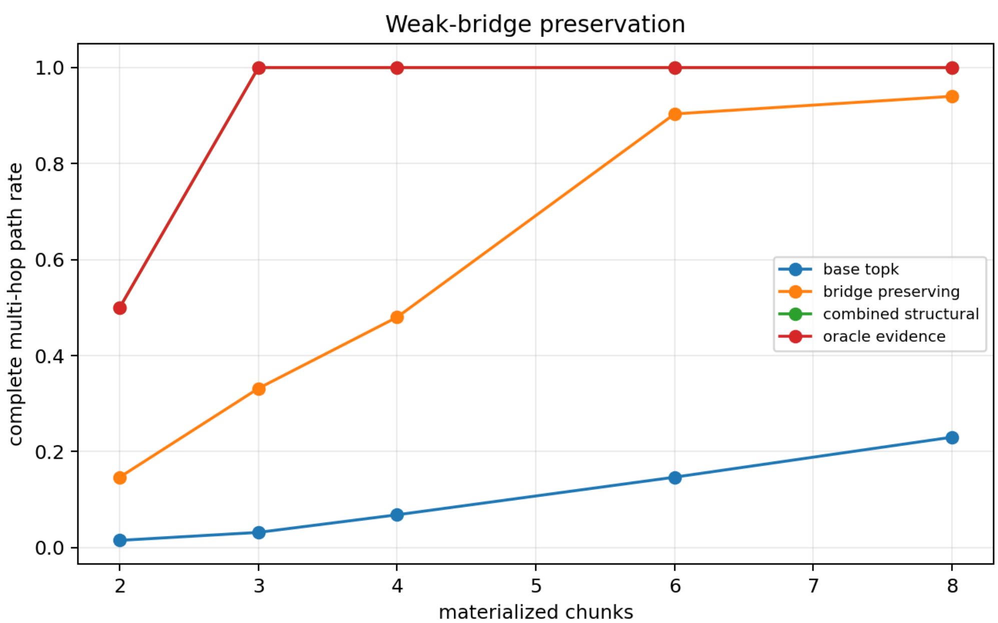

Figure 2: Complete evidence-path recovery across the budget sweep. Explicit bridge reservation prevents the weak-link failure induced by score-only selection.

Candidate removal provides exact causal utility in the controlled generator. Averaged across five held-out seeds, adding structural features raises Spearman from 0.486 to 0.528 and R^2 from 0.429 to 0.891. The result supports H1 only inside the controlled mechanism.

In the frozen-Qwen audit, surface-control Spearman is -0.027 and the structural model reaches only -0.005. Corresponding R^2 values are -0.352 and -0.348. The improvement is negligible and both models underperform the test-set mean. Retaining this negative result prevents observational graph richness from being mistaken for predictive utility.

Dataset	Budget	Base	Bridge	Paired delta (95% CI)
HOTPOTQA	64	-5.308	-4.223	+1.086 [-0.688, +2.943]
HOTPOTQA	128	-4.446	-4.033	+0.413 [-0.497, +1.710]
HOTPOTQA	192	-4.157	-3.517	+0.640 [-0.210, +1.846]
QASPER	64	-4.141	-3.599	+0.542 [-0.021, +1.363]
QASPER	128	-3.750	-3.739	+0.011 [-0.003, +0.035]
QASPER	192	-3.171	-2.668	+0.503 [-0.042, +1.550]

Table 3: Frozen-Qwen natural candidate intervention. Quality is mean answer-token log probability; all budgets use measured native K/V.

Bridge preservation improves the mean at all six natural operating points. On HotpotQA its paired answer-logprob deltas are +1.086, +0.413, and +0.640 at 64, 128, and 192 tokens. QASPER deltas are +0.542, +0.011, and +0.503. The pattern is directionally consistent, and evidence recall generally rises, but all paired intervals cross zero. S5 does not inherit the synthetic ceiling: its deltas are negative in five of six cells. On removal utility, adding structure changes HotpotQA Spearman from -0.385 to 0.021 while worsening R^2 from -0.135 to -5.391; on QASPER it changes Spearman from 0.161 to -0.028 and R^2 from -0.265 to -1.661. These small diagnostic removal subsets do not support incremental causal prediction.

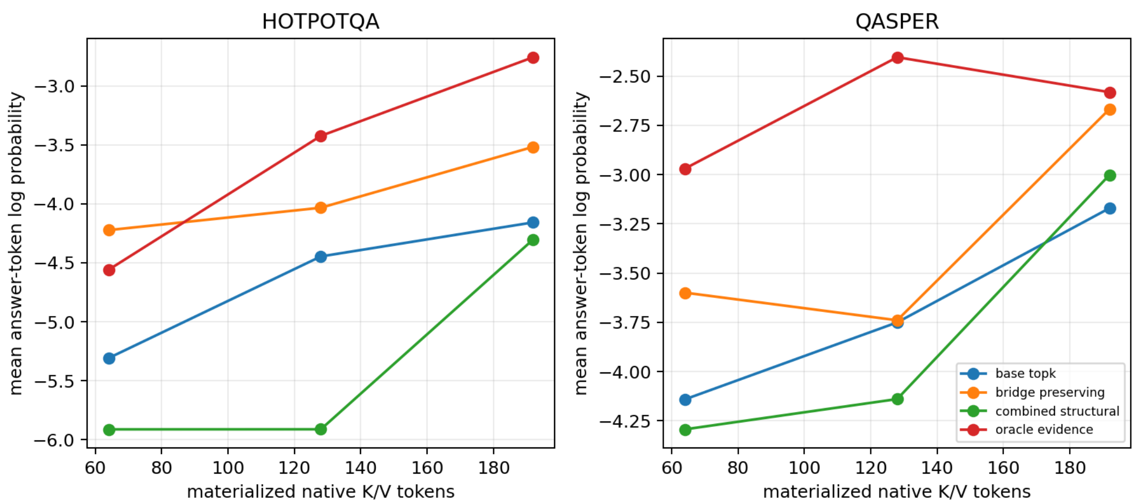

Figure 3: Natural frozen-Qwen quality versus measured native K/V tokens. Bridge means exceed B0 throughout, but paired uncertainty includes zero; S5 fails to generalize reliably.

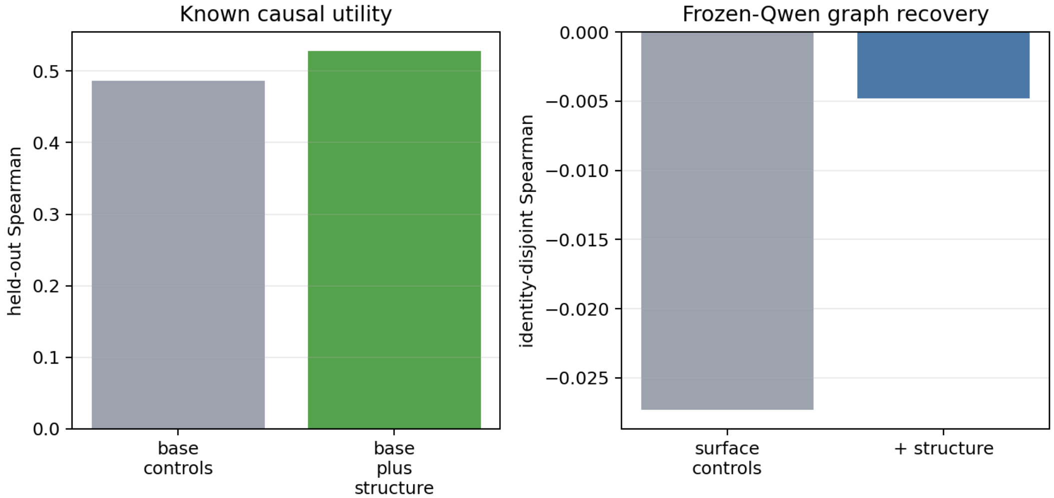

Figure 4: Left: controlled candidate-removal utility. Right: observational pretrained graph recovery after identity-disjoint fitting. Structural prediction is positive only in the known synthetic mechanism.

9 Series-Level Interpretation

The four-paper sequence supports a coherent but deliberately narrowing account. Paper 0 defines an evidence ladder in which recurrence and geometry are observations, interventions establish causal relevance, and persistence requires held-out utility. Paper 0.5 then shows why that separation matters: n-gram component responsibility changes with transformation and intervention, while motif specificity is unrelated to SA causal contribution ($\rho = -0.009$). Paper 0.6 generalizes the discrepancy to semantics: hierarchy-neighbor recovery exceeds permutation (0.477 versus 0.254), yet semantic motif specificity is negative (-0.012), and parent/root resolution recruits different component mixtures.

Paper 1 supplies the first direct sparse-control test. It shows that task-conditioned aggregate structure can matter: reserving a weak cross-community bridge changes the controlled frontier, whereas monotone score sharpening cannot alter exact top-k membership. The natural intervention adds a weaker result: bridge means move in the same favorable direction on both tasks, but uncertainty, S5 failure, and removal-prediction failure prevent a gate pass. The combined interpretation is therefore not that attention motifs become latent symbols or that abstraction rises monotonically through depth. Recurrent computation exposes candidate summaries whose utility is conditional on task, budget, and outcome; favorable averages are not yet a persistent-learning signal.

This distinction also explains the current stop decision. The candidate-level experiment closes the instrumentation gap but not the statistical one. The next step is a preregistered, larger multi-seed replication of the same frozen intervention, not a more expressive learner that could absorb the small validation slice.

10 Systems Discipline

All selectors enforce exact cardinality; both artifact families report zero budget violations. Natural traces contain candidate, selection, and evidence IDs and explicitly record that full attention is not retained. Unlike the controlled accounting estimate, the natural study separately records selector time, native K/V prefill time, end-to-end answer scoring, materialized tensor bytes, and device allocation. These research-code timings are not production serving claims.

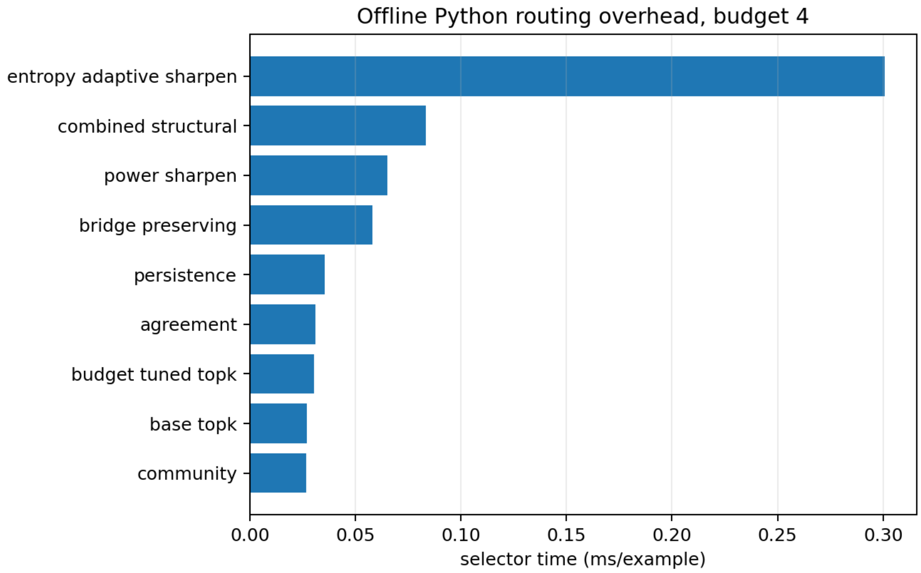

Figure 5: Research-code selector overhead at four chunks. Times measure CPU Python selection only and are not end-to-end PRA latency.

The implementation is opt-in. With NSC disabled, the adapter boundary calls exact base top-k and changes neither scores nor candidate membership. Deterministic component labels, bridge scores, sharpening monotonicity/normalization, budget enforcement, trace compactness, and disabled parity are covered by tests.

11 Decision Gate and Falsification

The controlled suite supports H1–H3 only under its explicit generator. The natural intervention satisfies the engineering requirements—candidate regeneration, identity separation, actual K/V materialization, equal budgets, task quality, and removal diagnostics—but does not satisfy reproducibility. All bridge means are favorable, yet every paired interval includes zero; S5 and held-out removal prediction fail. Online prototypes or adapters therefore remain blocked. A larger multi-seed replication may test the directional bridge pattern once, but no test-set tuning or learner expansion is justified.

12 Reproducibility and Provenance

The exact command is

```
python -m cl.experiments.paper1_structural_control
PYTHONPATH=src python -m cl.experiments.paper1_natural_gate
```

The manifests record resolved configuration, exact model/tokenizer revision, identities, cache hashes, package versions, device/dtype, and artifact hashes. The natural artifact adds 372 candidate rows, 480 selector rows, 48 removal rows, paired intervals, compact traces, measured systems fields, a generated table, and a figure.

13 Limitations

The controlled generator encodes the mechanism that structural features are designed to detect; its perfect S5 ceiling is diagnostic. The natural study is also deliberately small: eight held-out identities per dataset, one seed, twelve fixed candidates, coarse paragraph-derived evidence labels, one removal example per split, one model size, and answer likelihood rather than full answer generation. Its channel groups summarize representation subspaces rather than literal attention heads. Native K/V and latency are measured, but on Apple MPS research code rather than an optimized serving stack. These constraints make the favorable bridge direction a replication target, not a gate pass.

14 Conclusion

Transient graph structure is an effective frozen sparse-control signal when weak bridges are known to be causal. On natural HotpotQA and QASPER candidates, bridge reservation improves the mean quality–materialization frontier consistently but inconclusively; intervals include zero, S5 fails, and removal prediction remains out of sample. Paper 1 therefore contributes a reusable selector/trace/materialization framework, a positive controlled mechanism test, a suggestive natural direction, and a negative decision: do not consolidate without larger reproducible natural utility.

References

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